

Impact of Soundfont and Transposition on Chroma Features

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Question/Interest



- Do predicted Chroma Features vary when a piece of music is transposed and played on different instruments?
 - Fascinating that harmony and tonality are perceived differently based on instrument and key
 - Explore how digital tools analyze music
 - Is there a interaction between soundfont and transposition?

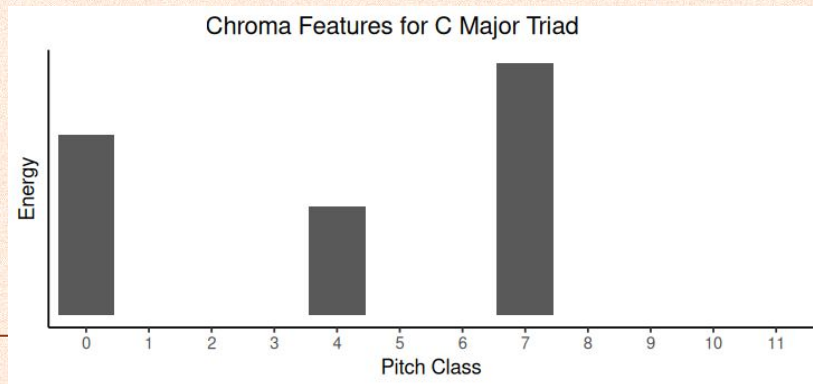
Methods: Files

- Used Chopin's 24 preludes, op.28
 - First 8 measures of the piece in each MIDI file
- Midi files were synthesized using 6 distinct instruments/soundfonts (grand piano, flute, guitar, harpsichord, honky tonk piano, organ)
- Transposed each piece by a range of 24 different semitones, ranging from up an octave to down an octave (-12 to 12)
- Total of 3600 audio files

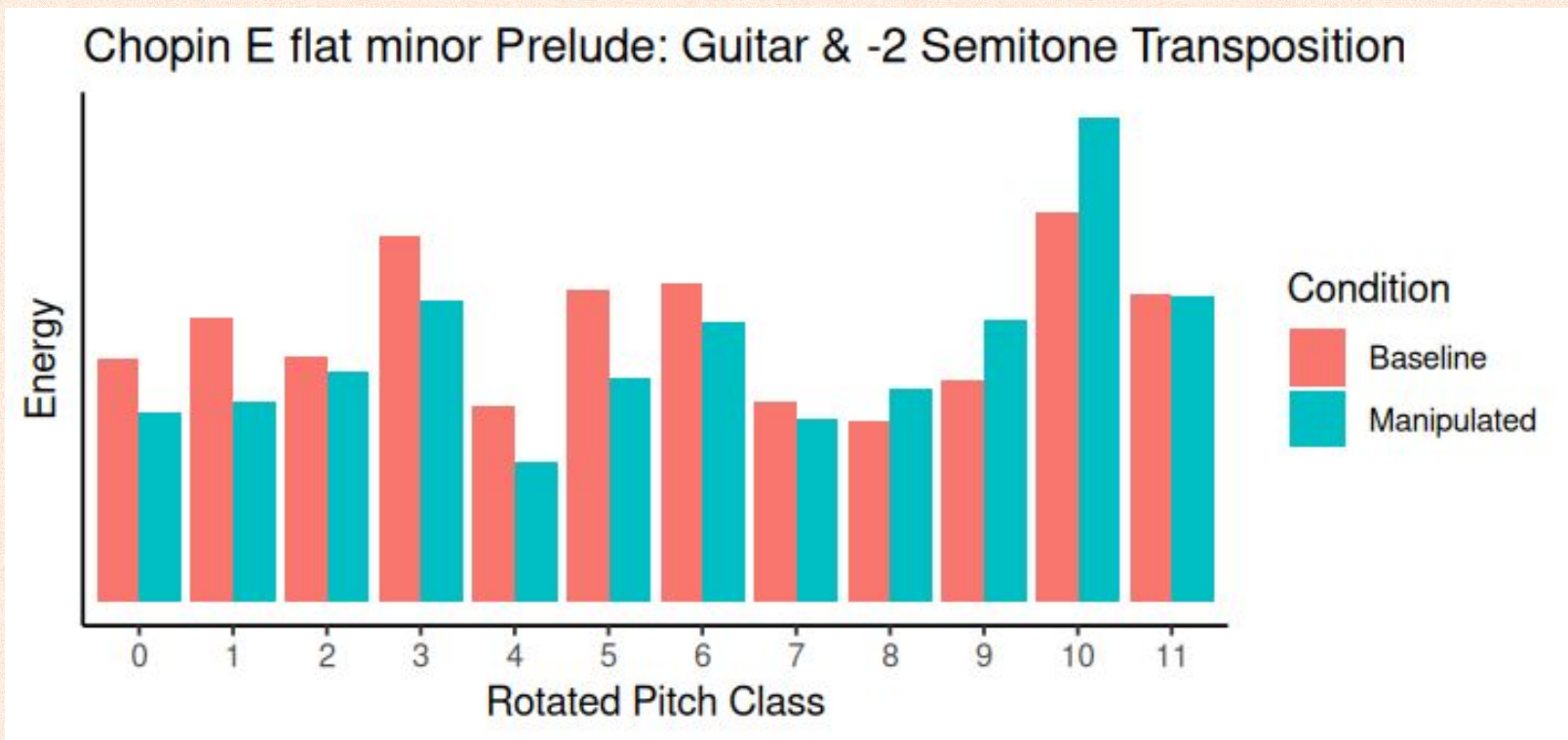
Methods: Feature Extraction

Chroma Features: A feature that represents the amount of energy at each of the 12 pitch classes, regardless of octave.

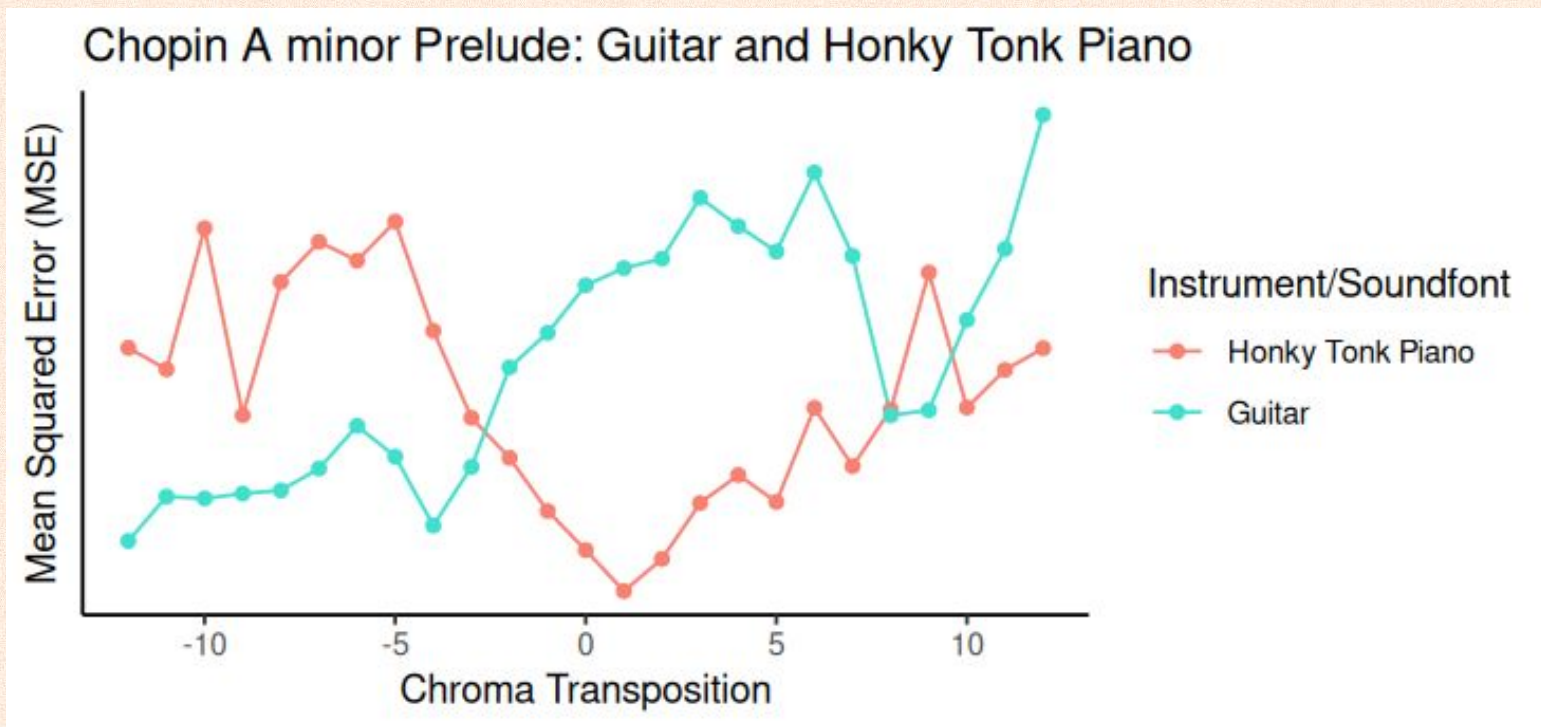
- Predicted from audio file by an algorithm using Librosa Constant-Q transform with default settings
- Useful for identifying musical elements such as chords and scales



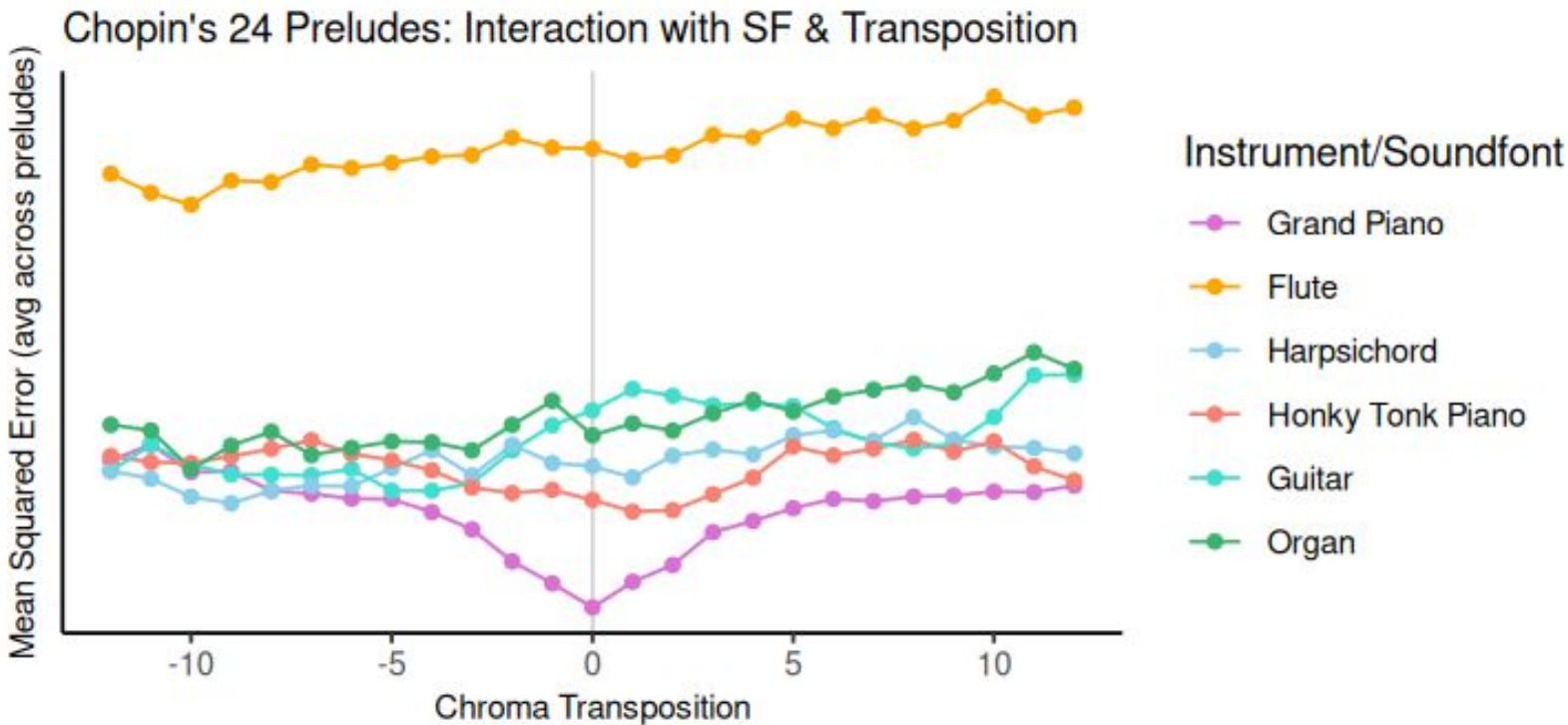
Results: Baseline vs. Manipulated



Results: Comparing Error Across Manipulations



Results: All Data



0 on x axis -> sole contribution of soundfonts

Purple (grand piano) instrument -> sole contribution of semitone transposition



Conclusions

- Linear Relationships: negative transpositions have lower MSE
- Sporadic Changes: MSE changes drastically between small changes in semitone transpositions
- Flute soundfont has a drastically higher MSE than other soundfonts